# Weekly Report 6
# Automated Biometric System for Individual Re-Identification of Mugger Crocodile

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## 1 Introduction

The goal of this project is to develop an automated biometric system capable of recognizing individual Mugger crocodiles using aerial imagery.

As per our plan from last week, we refined the two-step re-identification pipeline. We focused on the following:

- Improving YOLO-based dorsal scute detection and extending it to auto-annotation the dataset with confidence-based filtering.
- Using metric learning on a ResNet50-based model with triplet loss and balanced batch sampling for feature extraction and many-to-one identity mapping.

## 2 Overall method

The complete system operates as follows:

- Input crocodile images.
- Detect dorsal scute regions using YOLOv8.
- Select the highest-confidence bounding box per image.
- Extract embeddings using Resnet50.
- Perform identity matching using distance-based comparison in embedding space.

# 3 Improved Detection using YOLOv8

## 3.1 Model and Training

- Model: YOLOv8 (fine-tuned on annotated dorsal scute dataset)
- Training performed on a subset of manually annotated images.
- Single-class detection (scute localization only).

## 3.2 Key Improvements

- Lowered confidence threshold during inference to capture difficult detections.
- Increased input resolution to improve localization accuracy.
- Introduced selection of only the highest-confidence bounding box per image.

## 3.3 Auto-Annotation

- YOLO model used to automatically annotate the full dataset.
- Confidence-based filtering applied to remove unreliable detections.

# 4 Cropping and Preprocessing

Initial cropping resulted in multiple noisy and inconsistent regions.
We improved it by doing:

- Single crop per image (highest confidence bounding box).
- Bounding box padding to include contextual information.
- Removal of low-confidence and extremely small detections.
- Aspect ratio preservation using resize with padding (letterboxing).

# 5 Re-Identification Pipeline

- Cropped dorsal scute images are passed to a deep embedding model.
- The task is formulated as a metric learning problem rather than classification.
- The model learns an embedding space where similar identities are closer.

# 6 Model Architecture and Training

## 6.1 Feature Extraction

For feature extraction, we used ResNet-50. Since we were not doing simple classification, we removed the final fully connected layer and replaced it with an embedding head. It projects the image features into a 128-dimensional vector space. We also applied L2 normalization to these embeddings so that we can use simple Euclidean distance to tell if two scute patterns belong to the same crocodile.

## 6.2 Learning Strategy

We trained the model using Triplet Loss. The idea was to pull the "Anchor" ($a$) and "Positive" ($p$) images together while pushing the "Negative" ($n$) image away by at least a margin of $\alpha$.

$$\|f(a) - f(p)\|^2 + \alpha < \|f(a) - f(n)\|^2 \tag{1}$$

## 6.3 Batch Sampling and Triplet Combination

Random sampling was ineffective as batches did not have sufficient same-identity samples. To address this, we used Balanced Batch Sampling, ensuring multiple identities with multiple samples per identity in each batch.

For triplet selection:

- **Random Mining:** Inefficient, as most triplets were too easy.
- **Hard Mining:** Caused instability due to noisy or incorrect samples.
- **Semi-Hard Mining:** Most effective, selecting moderately difficult negatives and enabling stable training.

# 7 Results

The detailed results are summarized in Table 1.

Table 1: Model Evaluation Results: Predictions and Confidence Scores

| Image | Test ID | Predicted | Conf |
|---|---|---|---|
| class_145 | 145 | 109 | 0.7643 |
| class_145 | 145 | 109 | 0.7531 |
| class_146 | 146 | 109 | 0.7039 |
| class_146 | 146 | 110 | 0.8712 |
| class_147 | 147 | 107 | 0.5988 |
| class_147 | 147 | 107 | 0.5592 |
| class_148 | 148 | 109 | 0.5837 |
| class_148 | 148 | 110 | 0.5305 |
| class_149 | 149 | 110 | 0.8468 |
| class_149 | 149 | 109 | 0.7523 |
| class_150 | 150 | 110 | 0.7376 |
| class_150 | 150 | 110 | 0.6186 |
| class_151 | 151 | 109 | 0.6379 |
| class_151 | 151 | 110 | 0.5656 |
| class_152 | 152 | 110 | 0.7671 |
| class_152 | 152 | 109 | 0.7443 |
| class_153 | 153 | 109 | 0.8357 |
| class_153 | 153 | 109 | 0.8743 |
| class_154 | 154 | 109 | 0.7349 |
| class_154 | 154 | 109 | 0.9281 |
| class_155 | 155 | 109 | 0.7013 |
| class_155 | 155 | 109 | 0.8833 |
| class_156 | 156 | 109 | 0.6829 |
| class_156 | 156 | 110 | 0.7982 |
| class_157 | 157 | 110 | 0.7825 |
| class_157 | 157 | 109 | 0.9326 |
| class_158 | 158 | 109 | 0.8268 |
| class_158 | 158 | 110 | 0.8247 |
| class_159 | 159 | 109 | 0.8681 |
| class_159 | 159 | 110 | 0.5400 |
| class_160 | 160 | 110 | 0.8585 |
| class_160 | 160 | 110 | 0.6620 |

# 8 Future Work

- Distance-based retrieval system for identity matching.
- Improved domain generalization techniques.
- Exploration of alternative losses contrastive loss.